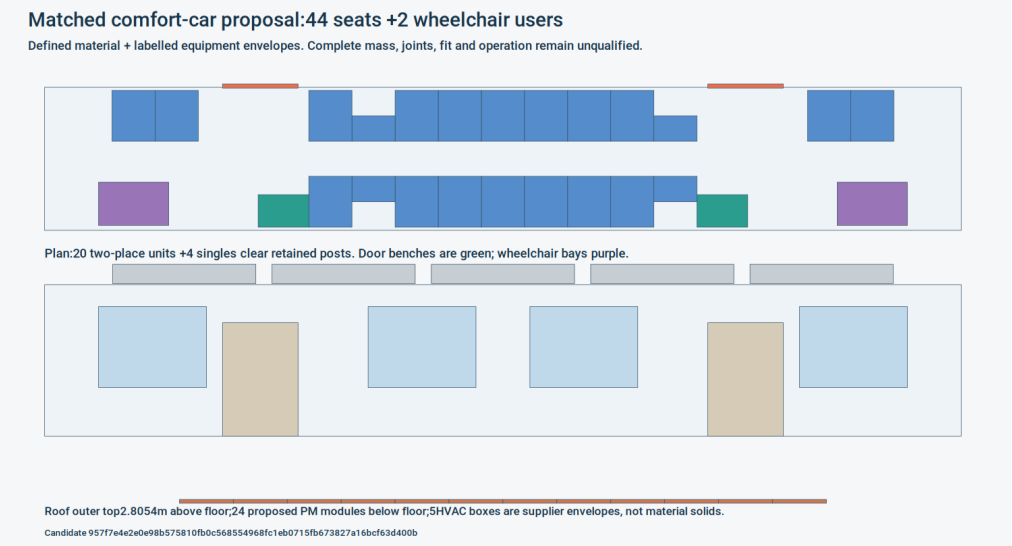
OpenSourceRail

Comfort-first development design

Simple vehicles. Repeated civil works. Frequent comfortable journeys.

Current repository design basis - 26 October 2026



Exact selected material geometry; equipment and unlocated allocations are not a complete installed assembly.

Six 17 m cars, each with 44 seats plus two wheelchair places and no planned standing. All magnets and suspension controls remain onboard. A concrete beam with passive steel facings keeps the ordinary guideway simple; station winding groups provide launch and controlled arrival.

Selected for development. Stable physical operation, complete installed mass, manufacturing release and operating release have not been established.

One configuration across the repository

The current selector, engineering dossier, 266 city calculations, delivery records, Workbench and this reader use the same physical identities. Historical evidence remains pinned to its actual source revision.

1. Ethos and system architecture

OpenSourceRail aims for low whole-life cost and useful passenger throughput through simple, repairable components and local manufacturing. The objective is a coherent railway, rather than the lightest individual part or an isolated passing simulation.

Location	Selected approach
Train	Permanent magnets, fast support/guidance control, sensors, essential power, passenger services and independently retained equipment.
Ordinary guideway	Repeated concrete beams and passive steel facings with attachment/alignment details; no permanent magnets or powered suspension equipment along ordinary spans.
Stations	Finite launch/receiving windings, routed phase/current resources, passenger platforms and admission controls.
Network	Fixed lines and transfers; explicit corridor, station, dwell, fleet and headway hypotheses.

Vehicle weight, support force, coil heating, cooling, structural deformation, guideway reactions, station propulsion and operating costs must close together. Numerical convergence alone is not feasibility.

Comfort is an explicit service choice

The normal cap is 46 people per car: 44 fixed seats and two wheelchair places, with zero planned standing. This reduces normal passenger load and improves seating availability. Enforceable admission and compatible accessible stations are required; accidental/crowd-load and evacuation cases remain in verification.

Six cars carry 276 normal passengers. At a 120-second interval, 30 departures/hour give 8,280 passengers/hour/direction. The former 14,400 benchmark is not achieved by this seating policy. Mean waiting is 60 seconds only for regular, unsaturated service.

2. Vehicle, structure and mass

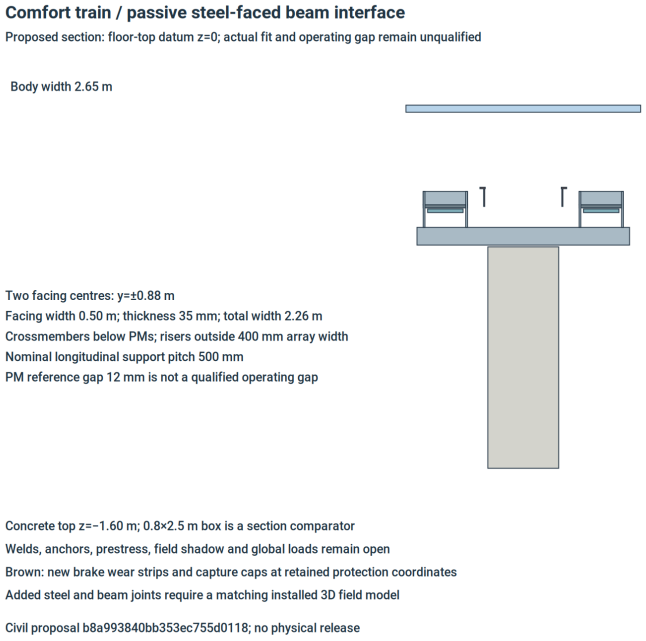
Subsystem	Selected development basis
Body/frame	Aluminium extrusion frame with retained GFRP side/end structure. A fully redesigned aluminium-only body is not established.
Floor	Aluminium skins/H80 core; selected counted floor stock 530.538 kg/car.
Roof	1.2 mm aluminium skins, 45 mm H60 core and perimeter bearings on retained ribs; counted roof stock 495.369 kg/car. Added crossrail alternative was rejected for interference.
Seats	20 paired benches plus four singles fit around retained structure; wall seating at door zones. Counted seat stock 258.294 kg within the retained interior account.
Doors/windows	Two passenger doors on one side. Revised 61 mm door leaves; supplier operator reference remains unqualified. Retained laminated glass; plastic alternatives need retention, stiffness, optics and release evidence.
Equipment	HVAC, batteries, wiring, controls, cooling, joints and protection must be located and weighed. Envelopes and accounting allocations are not invented material solids.

Mass per car	kg	Meaning
Accounted tare	17,356.684	Material, references and retained allocations
46-person capped load	20,806.684	75 kg/person declared hypothesis
160-person control	29,356.684	Retained load sensitivity, not the normal capacity
Complete installed tare/CG/inertia	Unknown	No complete installed mass properties claimed

Native CAD/STEP export and reopen established 20,776 material primitives and 10,198.351 kg of defined material. That is a material subset, not the complete equipped tare. Full joined-body structure, modes, fatigue and swept assembly/maintenance clearances remain open.

3. Suspension, gap and control

The selected onboard NdFeB Halbach assembly improves local force per kilogram relative to earlier ferrite research. It does not yet provide an accepted installed multiaxis force/current/temperature map, controllable equilibrium or stable hover.



Matched selected civil interface. The 12 mm coupon drawing gap is a reference pose, not an operating gap or boarding-height datum.

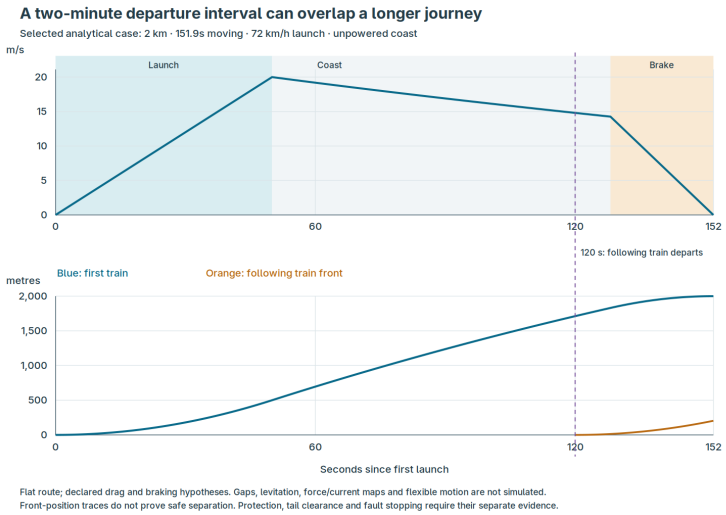
The lighter car has excess upward force at the retained 12 mm zero-current sample. Wider-gap samples require a different relative vehicle/facing position. Useful lift, lateral/rotational response, control current authority, temperature, damping and platform height must be solved together.

A smaller payload reduces required support force, but it does not by itself stabilize attractive magnetic suspension. Full-load reserve, force derivatives, fast control, essential power and mechanical retention remain necessary.

Electropermanent magnets may offer slow load trimming; they remain a separate research option. Fast stabilization, fault response and capture cannot be assumed from retained magnetic state or a scalar lift sample.

4. Service and finite station resources

Input/result	Selected hypothesis or calculation
Headway / dwell	120 s target / 35 s
Launch cap / station spacing	20 m/s (72 km/h) / 2,000 m prototype
Acceleration / deceleration	0.4 / 0.6 m/s2, flat route
Resistance	1,000 N/car + 1.8 v2 N/car; not measured drag
Runtime / arrival before braking	151.930 s / approximately 14.266 m/s
Station shared winding	800 m across launch and receipt; not 800 m for each separately
Sections / independent resources	64 addresses/direction at 12.5 m; at least two concurrent train phase/current branches/direction
Nominal resource interval	79.504 s; approximately 40.496 s below the 120 s target, not a safe-headway qualification



Analytical acceleration, drag-dependent coasting and braking for the selected accounted mass. This is not a physically levitating trip.

The winding library is 350/450/600/800/900/1,000 m. Each retained route leg selects its shortest full-engagement option. Terminal motors, overlapping intervals, phase handoff, actual voltage/current/thermal limits and safe stopping remain unqualified. The earlier 54 km/h/450 m prototype is an explicitly named comparison.

5. Baghdad and all-city recalculation

Baghdad retained-stop proposal	Current result
Fixed lines / added passenger stops	14 / 0
Active + spare + cold reserve	791 + 101 + 14
Trainsets / cars	906 / 5,436
Directional station winding	390,500 m
Library counts	331 x 800 m; 133 x 900 m; 6 x 1,000 m
Motor interval overlaps	4; handoff/resource qualification open
Mainline powered union coverage	Approximately 36.743%
Actual CAPEX / OPEX / finance	Unknown for this matched configuration

Fleet is recalculated from actual analytical coast/brake legs, retained terminal-time hypotheses, 35-second dwell, a 120-second interval, 12% rounded-up spares and one cold reserve per line. This is a planning requirement, not a vehicle purchase commitment.

All 266 city geographies and stops are retained. The same six-car comfort vehicle replaces older city-specific formations in current calculations. Geography and underlying map assets remain available; this migration does not regenerate or erase city basemaps.

Cities with unreachable coasting legs or no passing station-library option retain explicit failures and unknown complete totals. An unavailable total is not zero. Actual auxiliary loads, terminal energy, conversion losses and installed prices remain unqualified.

Catalogue summary	Cities
cities_with_all_station_library_screens	266
cities_with_station_library_failures	0
cities_with_unreachable_events	0

6. Civil, electrical and thermal closure

The new interface uses passive steel facings under a 2.26 m-wide envelope. It requires counted spreaders, supports, attachments and brake/capture load paths beyond the older narrow box. A conservative prototype counts approximately 865.1 tonnes of steel per directional kilometre, before remaining prestress, reinforcement, bearings, piers and foundations. This is a redesign input, not an optimized low-cost route.

Vehicle and magnetic reactions must drive moving-load envelopes, deflection, facing alignment, prestress, piers, foundations, transport and erection. Costs must include the complete installed route and construction sequence.

Hot-weather duty is a gating issue

The selected roof core fails the retained unshaded 0.5 solar-absorptivity, 900 W/m2, 50 C case against its continuous 70 C limit. A measured durable reflective finish, shading or material change is required; its mass and cost cannot be silently omitted.

HVAC references do not establish delivered cooling capacity or COP at 50 C. Passenger ventilation, solar gains, compressor phases, redundancy, stationary hot-weather heat rejection, charging and essential-energy endurance remain open.

Capture, braking and recovery

The retained eight-shoe undamped capture model exceeds stated energy/force/compression bounds in required drop cases. Larger alternatives require real geometry, mass and tested force/stroke data. Independent clamp-brake force and wet/hot friction are unqualified. A stopped train outside powered zones needs a real recovery system; a time-shifted trajectory does not create restart thrust.

7. What the software evidence demonstrates

Six actual Rust controls now execute against the newly selected 72 km/h analytical profile. Nominal, journal and fault cases check protection logic and retained authority. Their virtual healthy gaps and assumed braking floors are protocol inputs, not a magnetic or brake model of the selected train.

The native car material assembly is unchanged and independently verified from its portable source-bound archive. The selected IFC exports the counted civil shell, two steel facings and five HVAC contexts, with native geometry/identity checks; it is not a complete civil or vehicle BIM.

Earlier ideal-force Project Chrono journeys are method benchmarks. The physical milestone remains an empty and capped-load six-car train establishing verified hover, traversing finite launch equipment, coasting 1.5-2 km and arriving under measured force, electrical, thermal and braking limits, followed by fault cases.

Current delivery records

The engineering BOM, manufacturing, procurement, assembly, civil and controls packages share selection, vehicle, body and magnet identities. Missing supplier quotes, tooling, labour, origin evidence, serials, availability dates, construction crews and releases remain unknown. These records are suitable for research and requests for evidence, not fabrication authorization.

Reproducibility

Use ./osr status, ./osr build, ./osr verify, ./osr test and ./osr book --check. The Workbench reads the same selected packet. Unsupported physical jobs hold explicitly. ./osr legacy verify checks retained evidence at its actual Git revision; historical regeneration belongs in an isolated checkout.

8. Engineering-readiness register

Gate	State	Remaining evidence
mass	UNKNOWN	Equipped mass, CG and inertia retain unlocated allocations/references and unresolved installation stock.
suspension	UNKNOWN	Zero-current component samples have unresolved contour disagreement; no installed3D multiaxis current map, equilibrium or damping proof.
control	UNKNOWN	Actual current authority, supply bandwidth, gap/tolerance envelope and essential-power fault response are unqualified.
HVAC-electrical	UNKNOWN	Actual50C capacity/COP/fresh air, compressor phases and complete bus/charging duty remain unqualified.
roof-hot-core	FAIL	The retained0.5solar-absorptivity/900Wm2/50C unshaded hypothesis exceeds the H60 continuous70C limit; measured finish/shading or material revision is required.
body-clearance	UNKNOWN	Revised roof on retained ribs,61mm doors, floor inserts, seat restraints and magnetic attachments need native swept fit and joined-body analysis.
civil	UNKNOWN	New2.26m steel-facing envelope and spreaders need anchors, concrete/rebar, prestress, moving loads, magnetic shadow and site evidence.
capture-energy	FAIL	The retained eight-shoe undamped linear source model exceeds the source force/compression bounds in the required drop cases; alternatives need counted geometry and measured force/stroke.
onboard-braking	UNKNOWN	Retained clamp stock has declared force/friction/thermal sensitivities, but actual wet/hot friction, independent deployment and guideway load paths are unqualified.
recovery	UNKNOWN	A stopped off-zone train needs located powered or winch recovery and essential energy. Contact-supported towing can exceed the retained rope bound; time shift is not restart thrust.
operations	UNKNOWN	A nominal120s resource screen cannot establish support, guaranteed braking, receiving-power-loss protection or actual dwell distribution.
economics	UNKNOWN	Trainset unit price, priced BOM coverage, labour, tooling, station power, fleet and whole-life costs are incomplete.

9. Next experiments and source records

Magnetics and control

Resolve equilibrium and multiaxis current/temperature maps for the exact installed geometry; confirm convergence, reserve, damping and essential-power faults.

Complete equipped vehicle

Locate and weigh all installed stock; qualify doors, glazing, seats, roof finish, HVAC and support hardpoints; derive full CG/inertia and reduced structural modes.

Retention and stopping

Revise and count capture hardware; test energy, force and stroke. Establish independent braking friction/thermal duty and recovery outside powered zones.

Coupled demonstration

Use the same accepted component maps and vehicle mass/structure in stationary hover, launch/coast/arrival and fault simulations. Retain no-survivor and failed cases.

Whole-system comparison

Reoptimize civil stock and station groups, realistic dwell and transfers, manufacturing resources and equal-service costs before selecting a construction plan.

Traceable repository locations

engineering/development-baseline.toml

docs/design-engine/current/README.md

docs/design-engine/current/manifest.json

docs/design-engine/current/cities.json

docs/design-engine/current/exports/manifest.json

docs/ROADMAP.md

docs/archives/pre-comfort-default/manifest.json

Source revision for the retained physical dossier: 578ee3950b872bf30fbf9e4207b2bda85951bdd7. The current selector is a new development-selection event; it does not rewrite earlier scientific acceptance dispositions.

Appendix: current city planning coverage

Each row refers to the same six-car 46-person/car and 120-second development target. Trainsets are analytical requirements; unknown means one or more complete-route inputs fail. The screen is only the declared analytical coast/station comparison. Actual operating qualification and prices remain open.

City	Lines	Trainsets	Winding km	Screen
Benguela	5	149	57.6	Conditional
Huambo	5	150	58.2	Conditional
Lobito	4	103	36.6	Conditional
Luanda	14	852	381.7	Conditional
Lubango	4	141	50.2	Conditional
Malanje	2	42	16.2	Conditional
Namibe	5	122	39.6	Conditional
Soyo	4	90	33.4	Conditional
Uige	1	28	9.6	Conditional
Bukavu	5	164	64.6	Conditional
Goma	5	135	45.4	Conditional
Kananga	3	72	25.6	Conditional
Kinshasa	14	1200	638.1	Conditional
Kisangani	3	83	32.8	Conditional
Lubumbashi	8	294	124.9	Conditional
Mbuji Mayi	6	223	89.9	Conditional
Eldoret	5	168	68.0	Conditional
Garissa	4	81	29.2	Conditional
Kakamega	4	112	39.6	Conditional
Kisii	4	81	24.6	Conditional
Kisumu	5	139	47.8	Conditional
Kitale	4	110	42.4	Conditional
Machakos	4	88	28.2	Conditional
Malindi	4	87	31.2	Conditional

City	Lines	Trainsets	Winding km	Screen
Meru Ke	4	89	31.4	Conditional
Mombasa	9	464	206.3	Conditional
Nairobi	14	1045	458.7	Conditional
Naivasha	4	109	42.8	Conditional
Nakuru	5	145	52.0	Conditional
Nyeri	4	98	32.8	Conditional
Thika	5	178	67.0	Conditional
Antananarivo	14	660	275.7	Conditional
Beira	4	120	44.8	Conditional
Chimoio	3	87	30.6	Conditional
Lichinga	2	41	15.4	Conditional
Maputo	9	367	151.0	Conditional
Nacala	4	123	43.4	Conditional
Nampula	5	135	46.0	Conditional
Pemba Mz	4	97	36.2	Conditional
Quelimane	1	24	8.0	Conditional
Tete	5	126	45.0	Conditional
Xai Xai	4	69	19.6	Conditional
Huye	5	124	41.2	Conditional
Kigali	9	346	135.7	Conditional
Rubavu	4	110	41.2	Conditional
Mogadishu	6	238	103.7	Conditional
Waw	3	55	18.8	Conditional
Arusha	4	150	60.0	Conditional
Dar Es Salaam	14	1015	450.2	Conditional
Dodoma	5	135	51.0	Conditional
Iringa	4	83	29.4	Conditional
Kigoma	4	85	27.6	Conditional

City	Lines	Trainsets	Winding km	Screen
Mbeya	5	135	44.0	Conditional
Morogoro	5	132	45.4	Conditional
Moshi	4	93	31.4	Conditional
Mwanza	9	366	154.6	Conditional
Shinyanga	4	101	34.6	Conditional
Songea	2	35	9.6	Conditional
Sumbawanga	3	50	16.8	Conditional
Tabora	4	74	25.2	Conditional
Tanga	5	131	44.4	Conditional
Zanzibar City	5	140	54.2	Conditional
Arua	4	103	32.0	Conditional
Entebbe	4	119	51.4	Conditional
Fort Portal	4	99	36.2	Conditional
Gulu	5	165	68.0	Conditional
Hoima	4	84	29.2	Conditional
Jinja	4	134	60.4	Conditional
Kampala	9	424	184.4	Conditional
Lira	4	118	43.6	Conditional
Masaka	4	91	30.2	Conditional
Mbale	5	111	40.4	Conditional
Mbarara	4	142	56.6	Conditional
Soroti	2	31	12.2	Conditional
Lusaka	12	536	230.4	Conditional
Lyon	9	526	219.7	Conditional
La Paz	9	429	185.8	Conditional
Cuenca	5	166	64.6	Conditional
San Salvador	9	473	206.1	Conditional
Agadir	5	206	80.6	Conditional

City	Lines	Trainsets	Winding km	Screen
Beni Mellal	4	85	24.6	Conditional
Fez	6	239	97.4	Conditional
Kenitra	4	150	56.0	Conditional
Khouribga	3	53	14.6	Conditional
Marrakech	9	403	167.0	Conditional
Meknes	5	118	39.8	Conditional
Nador	4	100	34.2	Conditional
Oujda	4	93	29.8	Conditional
Safi	4	109	39.2	Conditional
Tangier	8	365	160.6	Conditional
Tetouan	4	145	56.0	Conditional
El Obeid	5	134	49.2	Conditional
Kassala	4	75	25.4	Conditional
Khartoum	14	817	356.0	Conditional
Nyala	5	125	40.4	Conditional
Omdurman	9	544	250.4	Conditional
Port Sudan	4	99	34.0	Conditional
Tunis	8	449	191.1	Conditional
Bloemfontein	5	200	86.2	Conditional
Durban	14	828	368.0	Conditional
East London Za	5	159	62.6	Conditional
Nelspruit	4	99	32.8	Conditional
Polokwane	5	144	53.8	Conditional
Herat	4	114	39.2	Conditional
Jalalabad Af	5	127	40.4	Conditional
Kabul	11	492	207.0	Conditional
Kandahar	5	148	50.8	Conditional
Mazar E Sharif	5	167	65.8	Conditional

City	Lines	Trainsets	Winding km	Screen
Barisal	5	168	67.2	Conditional
Chittagong	12	699	306.0	Conditional
Comilla	5	165	68.2	Conditional
Gazipur	9	513	221.4	Conditional
Khulna	9	404	173.0	Conditional
Mymensingh	4	120	47.0	Conditional
Narayanganj	5	192	71.6	Conditional
Rajshahi	4	111	42.8	Conditional
Rangpur	5	127	45.0	Conditional
Sylhet	4	135	51.0	Conditional
Agra	5	254	111.2	Conditional
Bhopal	6	337	149.4	Conditional
Coimbatore	9	590	265.0	Conditional
Indore	8	615	272.0	Conditional
Jodhpur	5	255	109.6	Conditional
Kanpur	8	606	264.0	Conditional
Lucknow	8	602	270.1	Conditional
Madurai	6	372	160.4	Conditional
Meerut	4	233	96.6	Conditional
Patna	6	478	227.7	Conditional
Raipur	6	260	106.3	Conditional
Rajkot	5	198	81.8	Conditional
Ranchi	6	378	167.7	Conditional
Vadodara	6	265	113.3	Conditional
Varanasi	6	355	154.1	Conditional
Vijayawada	6	342	141.6	Conditional
Visakhapatnam	6	414	180.6	Conditional
Biratnagar	4	93	31.8	Conditional

City	Lines	Trainsets	Winding km	Screen
Kathmandu	9	435	185.5	Conditional
Pokhara	5	196	80.4	Conditional
Bahawalpur	5	127	43.0	Conditional
Faisalabad	9	347	139.1	Conditional
Gujranwala	8	351	149.0	Conditional
Hyderabad Pk	9	388	163.5	Conditional
Karachi	14	971	435.4	Conditional
Larkana	3	90	30.4	Conditional
Multan	8	258	102.1	Conditional
Peshawar	8	353	142.4	Conditional
Quetta	6	214	85.3	Conditional
Rahim Yar Khan	5	156	55.6	Conditional
Sheikhupura	2	42	15.8	Conditional
Sialkot	5	144	50.8	Conditional
Sukkur	4	118	42.4	Conditional
Colombo	14	662	283.3	Conditional
Galle	5	171	60.8	Conditional
Jaffna	5	153	57.0	Conditional
Kandy	5	184	71.4	Conditional
Phnom Penh	9	531	234.7	Conditional
Bandung	6	431	202.5	Conditional
Surabaya	9	572	269.6	Conditional
Vientiane	5	187	72.8	Conditional
Mandalay	9	433	180.0	Conditional
Yangon	14	975	439.3	Conditional
Davao	9	508	217.3	Conditional
Ouagadougou	9	412	175.1	Conditional
Bafoussam	5	164	61.2	Conditional

City	Lines	Trainsets	Winding km	Screen
Bamenda	5	124	43.2	Conditional
Bertoua	3	62	17.8	Conditional
Douala	8	390	161.6	Conditional
Edea	1	21	7.2	Conditional
Garoua	4	94	31.8	Conditional
Kumba	4	111	38.2	Conditional
Maroua	5	141	49.8	Conditional
Ngaoundere	4	76	23.2	Conditional
Yaounde	8	425	176.8	Conditional
Conakry	5	183	74.9	Conditional
Bamako	9	519	229.2	Conditional
Niamey	9	345	143.5	Conditional
Aba Ng	4	88	29.8	Conditional
Benin City	8	286	114.7	Conditional
Ibadan	8	273	116.3	Conditional
Ilorin	5	136	47.8	Conditional
Jos	5	144	54.6	Conditional
Kano	14	858	371.3	Conditional
Maiduguri	8	340	145.5	Conditional
Onitsha	8	376	155.0	Conditional
Port Harcourt	8	363	155.5	Conditional
Uyo	4	94	34.2	Conditional
Dakar	9	481	218.6	Conditional
Arish	2	39	11.2	Conditional
Asyut	4	139	53.2	Conditional
Beni Suef	4	95	34.0	Conditional
Damanhur	5	136	47.2	Conditional
Damietta	5	184	74.8	Conditional

City	Lines	Trainsets	Winding km	Screen
Fayoum	5	164	59.8	Conditional
Hurghada	4	95	32.6	Conditional
Ismailia	4	134	49.0	Conditional
Kafr El Sheikh	4	94	33.4	Conditional
Luxor	5	148	57.2	Conditional
Mahalla	4	106	39.0	Conditional
Mansoura Eg	4	127	48.0	Conditional
Minya	4	132	49.6	Conditional
Port Said	4	83	29.6	Conditional
Qena	5	141	54.6	Conditional
Sohag	4	116	40.8	Conditional
Suez	5	160	58.8	Conditional
Tanta	5	184	69.2	Conditional
Zagazig	4	109	38.2	Conditional
Amarah	5	132	45.2	Conditional
Baghdad	14	906	390.5	Conditional
Baqubah	5	169	63.6	Conditional
Basra	11	598	248.1	Conditional
Diwaniyah	5	144	54.4	Conditional
Duhok	4	146	56.2	Conditional
Erbil	8	327	131.8	Conditional
Fallujah	5	140	49.6	Conditional
Hillah	5	165	58.0	Conditional
Karbala	9	336	137.6	Conditional
Kirkuk	8	305	119.2	Conditional
Kut	5	127	42.0	Conditional
Mosul	9	404	167.1	Conditional
Najaf	9	372	154.0	Conditional

City	Lines	Trainsets	Winding km	Screen
Nasiriyah	5	136	48.0	Conditional
Ramadi	5	130	46.0	Conditional
Samawah	5	136	49.2	Conditional
Sulaymaniyah	6	213	84.4	Conditional
Amman	14	754	323.3	Conditional
Aqaba	4	101	41.0	Conditional
Irbid	5	127	45.6	Conditional
Zarqa	5	178	69.6	Conditional
Beirut	9	318	134.4	Conditional
Sidon	4	94	29.2	Conditional
Tripoli Lb	4	125	46.8	Conditional
Gaza City	4	89	28.8	Conditional
Hebron	4	157	64.0	Conditional
Nablus	5	182	68.4	Conditional
Abha	4	165	64.4	Conditional
Al Kharj	5	165	62.2	Conditional
Buraidah	5	175	67.0	Conditional
Dammam	9	513	220.2	Conditional
Hail	5	170	62.2	Conditional
Hofuf	5	188	69.6	Conditional
Jeddah	14	750	318.1	Conditional
Jizan	5	142	51.2	Conditional
Khamis Mushait	5	216	83.8	Conditional
Mecca	9	431	176.7	Conditional
Medina	9	361	148.6	Conditional
Najran	5	157	56.4	Conditional
Tabuk	5	164	61.4	Conditional
Taif	5	158	54.6	Conditional

City	Lines	Trainsets	Winding km	Screen
Aleppo	9	382	168.8	Conditional
Damascus	9	400	158.4	Conditional
Deir Ez Zor	5	151	56.6	Conditional
Hama	5	151	56.2	Conditional
Homs	4	108	38.6	Conditional
Idlib	4	94	32.2	Conditional
Latakia	5	122	44.6	Conditional
Raqqa	4	112	40.6	Conditional
Tartus	4	84	28.0	Conditional
Aden	5	135	48.6	Conditional
Dhamar	4	92	37.0	Conditional
Hodeidah	4	83	24.0	Conditional
Ibb	5	140	53.0	Conditional
Lahij	4	83	29.6	Conditional
Mukalla	5	166	61.0	Conditional
Sanaa	11	543	238.1	Conditional
Sayun	4	76	24.2	Conditional
Taiz	5	130	44.7	Conditional